

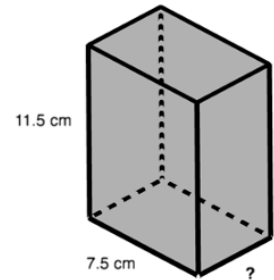
Grade 6 Accelerated  
Unit 8  
Lesson 3 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

A rectangular prism is shown with measurements in centimeters (cm).

1. What is the area of the front face of this prism? Include units.

$$11.5 \times 7.5 = 86.25 \text{ cm}^2$$



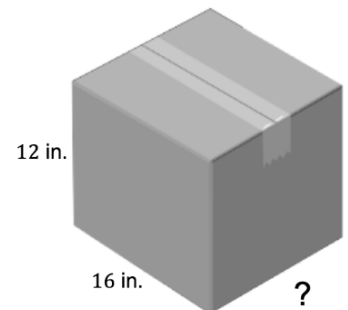
2. If the volume of the box is 345 cubic centimeters ( $\text{cm}^3$ ), what is the value of the missing dimension of the box?

$$\frac{345}{86.25} = 4, 4 \text{ cm}$$

Stefani is packing moving boxes. The boxes have a length of 16 inches (in.) and a height of 12 in., but she cannot recall the height of the box.

3. Determine the area of the front face of the box. Include units.

$$12 \times 16 = 192 \text{ in}^2.$$

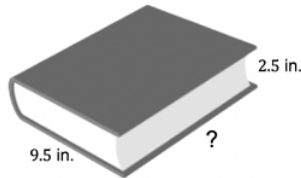


4. If the volume of the box is 2,304 cubic inches ( $\text{in}^3$ ), what is the width of the box? Include units.

$$\frac{2304}{192} = 12 \text{ in.}$$

Jeremiah was trying to find the missing dimension of the journal shown. He made a mistake when solving for the width of the box, but he cannot find it. His work is shown.

Volume = cubic in.



| Jeremiah's work                                                                                                                                 |                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $\begin{array}{r} 9.5 \\ \times 2.5 \\ \hline 455 \\ + 1900 \\ \hline 2355 \end{array}$ <p><math>2355 \rightarrow 23.55 \text{ in}^2</math></p> | $285 \div 23.55 = 12.1$ <p><math>w = 12.1 \text{ in.}</math></p> |

5. Circle the mistake.
6. What should the width of the box be? Include units.

$$285 \div 23.75 \text{ in.} = 12 \text{ in.}$$

7. Write a note to Jeremiah on how to correct his error(s).  
**Don't forget to carry the tens values when finding products; you should have carried the 2 when multiplying  $5 \times 5$ .**
8. A box of crayons has a length of 1.5 inches (in.) and a height of 4 in. The box has a volume of 16.8 cubic inches (in<sup>3</sup>). What is the width of the box? Include units.

$$\frac{16.8}{(1.5 \times 4)} = \frac{16.8}{8} = 2.8 \text{ in.}$$

9. Josh bought a sandbox. He knows that the length is 3.5 feet (ft) and the width is 2.5 ft., but he can't remember the appropriate height. The volume needs to be 10.5 cubic feet (ft<sup>3</sup>). What is the height of the sandbox? Include units.

$$\frac{10.5}{(2.5 \times 3.5)} = \frac{10.5}{8.75} = 1.2 \text{ ft.}$$

10. Cara is measuring her suitcase before her trip. The case has a width of 8.75 inches (in.) and a height of 14 in. The volume of the case is 2,756.25 cubic inches (in<sup>3</sup>). Determine the length of the suitcase. Include units.

$$\frac{2756.25}{(8.75 \times 14)} = \frac{2756.25}{122.5} = 22.5 \text{ in.}$$

